

MANGROVE MOUNTAIN, Australia

WMO: 957740

Lat: 33.2894S

Lon: 151.2108E

Elev: 305

StdP: 97.71

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.2	4.3	-3.3	3.0	15.3	-1.7	3.4	12.6	8.6	13.9	7.4	13.3	1.5	310	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	34.0	20.6	31.5	20.2	29.4	19.8	23.2	29.3	22.3	27.6	21.5	26.2	3.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.9	25.1	20.9	16.1	23.9	20.2	15.4	23.0	70.7	29.4	67.1	27.7	64.2	26.4	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.2	5.4	DB	1.0	38.6	0.8	2.3	0.5	40.3	0.0	41.6	-0.4	42.9	-1.0	44.5	
			WB	0.3	25.3	0.6	1.1	-0.2	26.1	-0.5	26.7	-0.9	27.4	-1.3	28.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	21.9	21.4	19.8	17.0	13.6	11.3	10.6	11.7	14.5	16.8	18.7	20.5
	DBStd	4.85	3.29	2.99	2.74	2.55	2.31	1.98	2.05	2.51	3.17	3.58	3.62	3.36
	HDD10.0	36	0	0	0	0	2	9	16	9	1	0	0	0
	HDD18.3	1132	5	4	15	56	147	210	238	207	122	74	38	14
	CDD10.0	2392	369	319	303	209	113	48	35	61	137	211	261	326
	CDD18.3	448	116	90	60	15	1	0	0	1	8	27	49	82
	CDH23.3	3244	877	558	301	76	3	0	0	1	81	256	429	662
	CDH26.7	1204	383	204	79	13	0	0	0	0	17	78	166	264
(14) Wind	WSAvg	2.4	2.8	2.5	2.3	1.9	2.1	2.2	2.4	2.5	2.6	2.6	2.6	2.6
(15) Precipitation	PrecAvg	996	102	141	100	90	74	95	49	63	51	57	83	88
	PrecMax	1708	331	521	261	365	267	414	156	263	156	196	196	202
	PrecMin	641	12	15	7	3	9	16	1	0	1	9	5	27
	PrecStd	261	63	117	57	83	60	78	39	64	39	45	48	46
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	35.8	31.8	28.6	23.4	19.5	20.2	22.7	29.0	32.0	35.1	36.1
		MCWB	22.0	22.3	20.4	18.1	15.9	13.7	13.3	12.8	15.4	16.4	19.2	20.5
	2%	DB	33.9	32.0	28.8	25.4	21.2	17.5	17.7	20.4	25.8	28.8	31.1	32.6
		MCWB	21.5	21.6	20.3	17.6	15.1	12.9	11.5	12.3	14.4	16.6	19.1	20.0
	5%	DB	30.9	28.9	26.5	23.4	19.5	16.3	16.1	18.6	23.0	26.2	28.0	29.5
		MCWB	21.0	21.3	20.0	17.3	14.3	12.4	11.1	11.8	13.8	16.0	18.0	19.7
	10%	DB	28.1	26.5	24.5	21.6	18.1	15.2	14.8	16.9	20.7	23.6	25.3	26.9
		MCWB	20.8	20.6	19.2	16.7	13.9	11.9	10.6	10.9	13.3	15.6	18.1	19.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	24.6	23.1	20.3	18.0	15.4	15.3	15.3	17.4	19.7	22.3	23.3
		MCDB	32.1	31.4	27.8	24.6	19.8	17.3	17.9	20.1	23.7	25.4	29.4	30.3
	2%	WB	23.3	23.1	21.7	19.1	16.8	14.2	13.4	13.9	16.2	18.6	20.9	21.9
		MCDB	30.2	28.1	26.1	22.5	18.9	15.9	15.3	17.3	21.3	24.1	27.4	28.5
	5%	WB	22.3	22.2	20.9	18.2	15.7	13.6	12.3	12.9	15.3	17.6	19.7	20.9
		MCDB	28.0	26.3	24.4	21.2	18.1	15.2	14.4	16.6	20.0	22.8	25.2	26.5
	10%	WB	21.5	21.3	20.1	17.5	14.8	12.8	11.4	11.9	14.5	16.7	18.8	20.1
		MCDB	26.0	24.9	22.9	20.4	17.2	14.4	13.7	15.4	18.9	21.3	23.4	24.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	8.9	8.6	8.9	9.0	7.8	8.7	10.1	10.7	10.6	10.1	10.3
		MCDBR	15.9	13.7	12.2	11.4	11.0	9.4	10.5	12.2	14.1	15.2	15.5	15.3
	5% WB	MCWBR	5.7	5.4	5.2	5.4	5.8	5.7	6.0	6.2	6.0	6.1	5.7	5.6
		MCWBR	12.7	11.0	10.3	9.6	8.7	7.2	7.9	10.0	11.2	12.1	12.5	12.6
(40) Clear-Sky Solar Irradiance	taub		0.383	0.375	0.344	0.321	0.307	0.298	0.291	0.297	0.330	0.353	0.378	0.375
		taud	2.460	2.508	2.583	2.621	2.607	2.633	2.639	2.614	2.511	2.470	2.438	2.467
	Ebn at Noon		955	943	938	908	869	853	881	922	937	953	953	966
		Edh at Noon	119	109	95	81	73	67	70	80	100	112	121	119
(44) All-Sky Solar Radiation	RadAvg		6.47	5.63	4.78	3.87	3.06	2.44	2.85	3.76	4.96	5.69	6.07	6.60
	RadStd		0.46	0.49	0.42	0.33	0.23	0.21	0.22	0.23	0.27	0.51	0.56	0.56

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.40	-0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.45	N/A	N/A	+0.84	N/A	N/A	-2	-86	+190	+102

Nomenclature: See separate page